

FRE-GT-9743 Assignment 1

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Bond Forward Business

Bond forward is one of the simplest derivatives for an interest derivative desk. The payoff is:

$$V(T_s) = B(T_s; T_s, T_m) - K \quad (1)$$

where $B(T_s; \cdot, \cdot)$ stands for the bond price at settlement date T_s , and K is the strike agreed to make the contract at par at inception. According to arbitrage-free pricing theory, at any time $t \in [0, T_s]$, the mark-to-market (MTM) value of the contract is:

$$V(t) = df_{\text{csa}}(t, T_s)(B(t; T_s, T_m) - K) \quad (2)$$

where df_{csa} is the CSA discounting and $B(t; T_s, T_m)$ is time- t forward bond price.

You're the rates desk at bank A, a client comes to you to buy 1,000,000 1-year forward contract on a US Treasury 10-year.

Q1. What could be the motivation of the client to get into such a deal? What are the important parameters on the term sheet (contract)? On the settlement date T_s , what does bank A pay/deliver, what does client pay?

Ans: The client might want to lock the 10-yr US Treasury price, in case the price will increase and the future interest rate or expected yields goes down. Or maybe this is just his future asset allocation— he will buy bond in the future at a known price.

I think important parameters are interception date, settlement date, maturity date, and the strike. Because these all influence the value of the forward. Apart from these, the notional, settlement method and CSA terms are also important.

If it is physical settlement, bank will give client bond and the client pays K times quantities. If it settled in cash, it will be paid by bilateral netting from bank and client ($\max(B(T_s; T_s, T_m) - K, 0)$).

The most important quantity in (2) is $B(t; T_s, T_m)$, in particular, when $t = 0$. For those students who took option pricing, you can work out the pricing with change-of-numeraire formula (recall your final exam!). However, here, let us use a more practical approach that traders use. This is what Q2–Q3 about.

Q2. To avoid market fluctuation, at inception ($t = 0$), the trading desk will buy the spot bond at the current market price $B(0; 0, T_m)$ – why? To fund this bond position, how do we arrange internal trade with treasury desk and repo desk?

Ans: The desk buys at the interception so that the price is locked and the future market risk is eliminated. To fund for the bond, the bond is posted as collateral. They sell the bond to repo desk and receive cash $B(0;0,T_m)$ and agree to purchase at T_s at repo rate R .

Q3. Suppose the repo desk quote you a 1-year repo rate R , can you compute $B(0;T_s,T_m)$ via the following inputs

- Treasury 10-year bond coupon schedule;
- Term repo rate R
- Time-0 spot price $B(0;0,T_m)$.

Hints:

1. if you borrow \$1 from repo desk, how much you have to pay back at maturity?
2. typically, for this repo transaction, you will have to pass the bond as collateral to the repo desk. That means, from 0 to T_s , there could be some coupons paid to repo desk, would this decrease your debt?

Ans: Any coupon before T_s received can reduce debt. If we borrow $B(0;0,T_m)$ at time 0, at T_s we will pay $B(0;0,T_m)e^{RT_s}$, where R is repo rate. So at time 0 the price should be

$$B(0;T_s,T_m) = B(0;0,T_m)e^{RT_s} - \sum_{t_i \leq T_s} C_i e^{R(T_s-t_i)}$$

Q4. On any $t \in (0, T_s]$, how do you compute the MTM of the bond forward contract?

Ans: According to arbitrage-free pricing theory, at any time $t \in [0, T_s]$, the mark-to-market (MTM) value of the contract is:

$$V(t) = df_{csa}(t, T_s)(B(t;T_s,T_m) - K)$$

Let $P(t)$ be the dirty spot price of the bond at time t , $P(t) = B(t;0,T_m)$, the remaining coupon dates between t and T_s with coupon C_i . Let the repo accrual factor $A_t(u, v) = e^{R(v-u)}$. Plug into the MTM formula

$$V(t) = df_{csa}(t, T_s)(P(t)A_t(t, T_s) - \sum_{t < t_i \leq T_s} C_i A_t(t_i, T_s) - K)$$

Q5. If the desk follows Q2 and Q3, does desk expose to any major market risk? If not, that also means, desk makes no money? How would you convince your client so that you can get some revenue from this trade?

Hint: think about repo rate!

Ans: There will largely be no interest-rate (price) risk, because the forward exposure to the bond price is offset by holding the bond. And changes in yield move both the forward value and the spot bond value in the opposite way, so delta / duration risk is hedged.

However there are still some risks left, like repo rate. Repo funding rate can differ from: the client's implied financing, the model "risk-free" curve, or future realized repo levels. If repo becomes expensive (or the bond goes "special"), the desk's carry changes.

In reality the desk can earn revenue because there are non-zero spreads and constraints, mainly around funding (repo).

***Q5.** Can you relate Q2–Q3 to risk neutral pricing formulae you have learnt? I claim they're the same under certain assumptions.

Ans: Under the risk-neutral framework, the time- t forward price of the bond settling at T_s can be written as

$$B(t; T_s, T_m) = \frac{1}{P(t, T_s)} \mathbb{E}^{\mathbb{Q}} \left[B(T_s; T_s, T_m) + \sum_{t < t_i \leq T_s} C_i \middle| \mathcal{F}_t \right],$$

where $P(t, T_s)$ denotes the discount factor to T_s , and C_i are the deterministic coupon payments.

Choosing the T_s -zero-coupon bond as numeraire and working under the T_s -forward measure $\mathbb{Q}^{T_s}$, this becomes

$$B(t; T_s, T_m) = \mathbb{E}^{\mathbb{Q}^{T_s}} \left[B(T_s; T_s, T_m) + \sum_{t < t_i \leq T_s} \frac{C_i}{P(t_i, T_s)} \middle| \mathcal{F}_t \right].$$

Under the assumptions of deterministic interest rates, deterministic coupon cashflows, and repo funding at the risk-free rate, the expectation collapses and yields

$$B(t; T_s, T_m) = \frac{B(t; 0, T_m)}{P(t, T_s)} - \sum_{t < t_i \leq T_s} \frac{C_i P(t, t_i)}{P(t, T_s)}.$$

Identifying the repo accrual factor as

$$A(u, T_s) = \frac{1}{P(u, T_s)},$$

the above expression is identical to the cash-and-carry pricing formula derived in Q2–Q3:

$$B(t; T_s, T_m) = B(t; 0, T_m) A(t, T_s) - \sum_{t < t_i \leq T_s} C_i A(t_i, T_s).$$

Hence, the replication argument in Q2–Q3 and the risk-neutral pricing approach are equivalent under these assumptions.

Implementation of A Simple Interpolator

A 1D-interpolator can be viewed as a parametric function

$$f(\cdot; \theta = \{x_i, y_i\}'_s) \tag{3}$$

where x_i 's are given N -knots on x -axis, and y_i 's are the corresponding values. Given an arbitrary input x^* , the interpolated value is

$$y^* = f(x^*; \theta) \mapsto \mathbb{R}.$$

You're asked to build a simple piecewise constant left continuous (PCLC) interpolator with flat extrapolations by deriving from an abstract interpolator class. You will need to support the following functionalities:

- Interpolate a given $x^* \in \mathbb{R}$
- Return the gradient $\nabla_y f|_{x=x^*}$

- Integrate the interpolator over a region, i.e.,

$$I[f; l, u] = \int_l^u f(x; \theta) dx \quad (4)$$

- Return the gradient $\nabla_y I[f; l, u]$ of the integral.

You're asked to write you validation code to prove your gradient calculation is correct. This is done by a simple B&R (bump and reval).

Please follow the structure in the FixedIncomeLib.